2 Discrete Fourier Transform

Discrete Fourier Transform (DFT) is the Fourier representation of a finite length sequence, which itself is a sequence rather than a continuous function. It corresponds to sample equally spaced in the frequency of the Fourier transform of the signal. DFT will transform an N-point sequence $x(n), 0 \le n \le N-1$ into the N-point discrete Fourier domain samples $x(k), 0 \le k \le N-1$.

Representation of Periodic Sequences: Discrete Fourier Series (DFS)

Consider a sequence $\tilde{x}(n)$ that is periodic with period N, so that

$\tilde{x}(n) = \tilde{x}(n+k+N)$

A discrete time sequence $\tilde{x}(n)$ of period N can be expressed as a weighted sum of complex exponential sequences. The exponential form of the Fourier series for a periodic discrete time sequence is given by:

$$\tilde{x}(n) = \frac{1}{N}\sum_{k=0}^{N-1}\tilde{X}(k)e^{j\left(\frac{2\pi}{N}\right)kn} \qquad n = 0, 1, \ N-1 \qquad \ldots (i)$$

where the coefficients $\tilde{X}(k)$ are given by

$$\tilde{X}(k) = \sum_{n=0}^{N-1}\tilde{x}(n)\, e^{-j\left(\frac{2\pi}{N}\right)kn} \qquad \ldots (ii)$$

For convenience in notation, these equations will be generally written in terms of WN, which is defined as:

$$WN = e^{-j\left(\frac{2\pi}{N}\right)}$$

Thus, the DFS analysis and synthesis pair are expressed as:

$$\tilde{X}(k) = \sum_{n=0}^{N-1}\tilde{x}(n)\, W_N^{kn} \quad , \ k = 0.1, - - - N-1$$

$$\tilde{x}(n) = \frac{1}{N}\sum_{k=0}^{N-1}\tilde{X}(k)\, W_N^{-kn} \quad , \ n = 0.1, - - - N-1$$

Note: DFS coefficients are periodic with a period N, $\tilde{X}(k+N) = \tilde{X}(k)$

Solved Examples

1. Find the Discrete Fourier Series representation for the following sequence.

$$\tilde{x}(n) = \sum_{k=-\infty}^{\infty} x(n-10k)$$

Where, $x(n) = \begin{cases} 1 & 0 \le n < 5 \\ 0 & \text{else where} \end{cases}$

Solution:

$\hat{x}(n)$ is a periodic sequence with a period N = 10

The DFS coefficients are:

$$\tilde{X}(k) = \sum_{N=0}^{9}\tilde{x}(n)\, e^{-j\left(\frac{2\pi}{10}\right)kn}$$

$$= \sum_{n=0}^{4} e^{-j\left(\frac{2\pi}{10}\right)kn} = \frac{1-e^{-j\pi k}}{1-e^{\frac{-j\pi k}{5}}}$$

$$\tilde{X}(k) = \frac{1-e^{-j\pi k}}{1-e^{-j\frac{\pi k}{5}}}$$

$$\tilde{X}(k) = \begin{cases} 5 & k = 0 \\ \dfrac{2}{1-e^{\frac{-j\pi k}{5}}} & k \text{ is odd} \\ 0 & k \text{ is even} \end{cases}$$

Properties of Discrete Fourier Series

1. Linearity: If two periodic sequences $\hat{x}_1(n)$ and $\hat{x}_2(n)$ both with periods equal to N are combined according to linearity

$\tilde{x}_3(n) = a\tilde{x}_1(n) + b\tilde{x}_2(n)$

then the coefficients in the DFS representation of $\hat{x}_3(n)$ are given by:

$\tilde{X}_3(k) = a\tilde{X}_1(k) + b\tilde{X}_2(k)$

2. Shift of a sequence: If $\tilde{X}(k)$ are the DFS coefficients for $\hat{x}(n)$, then DFS coefficients for $\hat{y}(n) = \hat{x}(n-n_0)$ are:

$\tilde{Y}(k) = W_N^{kn_0}\tilde{X}(k)$

If $\tilde{X}(k)$ are the DFS coefficients for $\tilde{x}(n)$, then DFS coefficients for $\tilde{y}(n) = W_N^{k_0 n}\tilde{x}(n)$ are $\tilde{Y}(k) = \tilde{X}(k + k_0)$.

3. **Periodic convolution:** If $\tilde{h}(n)$ and $\tilde{x}(n)$ are periodic with a period N with DFS coefficients $\hat{H}(k)$ and $\hat{X}(k)$, respectively, then the periodic convolution of two sequences is written as:

$$\tilde{y}(n) = \tilde{h}(n) \otimes \tilde{x}(n)$$

$$= \sum_{m=0}^{N-1} \tilde{h}(m)\,\tilde{x}(n-m)$$

Then D.F.S. coefficient of $\hat{y}(n)$ are given by

$$\boxed{\tilde{Y}(k) = \tilde{H}(k)\tilde{X}(k)}$$

4. **Symmetry property:**

$$DFS\left\{\tilde{x}^*(n)\right\} = \tilde{X}(-k)$$

$$DFS\left\{\tilde{x}^*(-n)\right\} = \tilde{X}(k)$$

$$DFS\left\{Re\left[\tilde{x}(n)\right]\right\} = DFS\left[\frac{\tilde{x}(n) + \tilde{x}^*(n)}{2}\right]$$

$$= \frac{1}{2}\left[\tilde{X}(k) + \tilde{X}(-k)\right] = \tilde{X}_{cs}(k)$$

$\tilde{X}_{cs}(k) \Rightarrow$ Conjugate symmetric part of $\tilde{X}(k)$

$$\boxed{DFS\left\{Re\left[\tilde{x}(n)\right]\right\} = \frac{1}{2}[\tilde{X}(k) + \tilde{X}(-k)] = \tilde{X}_{CS}(k)}$$

$$DFS\left\{Im\left[\tilde{x}(n)\right]\right\} = DFS\left[\frac{\tilde{x}(n) - \tilde{x}^*(n)}{2}\right]$$

$$\boxed{DFS\left\{I_m\left[\tilde{x}(n)\right]\right\} = \frac{1}{2}\left[\tilde{x}(k) - \tilde{x}(-k)\right] = \tilde{X}_{CAS}(k)}$$

$\tilde{X}_{CAS}(k) \Rightarrow$ Conjugate anti-symmetric part of $\tilde{X}(k)$

$$\tilde{x}_{cs}(n) = \frac{1}{2}\left[\tilde{x}(n) + \tilde{x}^*(-n)\right]$$

$\tilde{x}_{cs}(n) \Rightarrow$ Conjugate symmetric part of $\tilde{x}(n)$

$$DFS\left[\tilde{x}_{CS}(n)\right] = DFS\frac{\left[\tilde{x}(n) + \tilde{x}^*(-n)\right]}{2}$$

$$= \frac{1}{2}\left[\tilde{X}(k) + \tilde{X}^*(k)\right] = Re\left[\tilde{X}(k)\right]$$

$$\boxed{DFS\left[\tilde{X}_{CS}(n)\right] = Re\left[\tilde{X}(k)\right]}$$

$$\tilde{x}_{cAS}(n) = \frac{1}{2}\left[\tilde{x}(n) - \tilde{x}^*(-n)\right]$$

$\tilde{x}_{CAS}(n) \Rightarrow$ Conjugate anti-symmetric part of $\tilde{x}(n)$

$$DFS\left[\tilde{x}_{CAS}(n)\right] = DFS\left[\frac{\tilde{x}(n) - \tilde{x}^*(-n)}{2}\right]$$

$$= \frac{1}{2}\left[\tilde{X}(k) - \tilde{X}(-k)\right] = I_m\left[\tilde{X}(k)\right]$$

$$\boxed{DFS\left[\tilde{x}_{CAS}(n)\right] = I_m\left[\tilde{X}(k)\right]}$$

Summary of the Properties of DFS Representation of Periodic Sequences

S. No.	Periodic Sequence (Period N)	DFS Coefficients
1.	$\tilde{x}(n)$	$\tilde{X}(k)$ periodic with period N
2.	$\tilde{x}_1(n), \tilde{x}_2(n)$	$\tilde{X}_1(k), \tilde{X}_2(k)$ periodic with period N
3.	Linearity: $a\tilde{x}_1(n) + b\tilde{x}_2(n)$	$a\tilde{X}_1(k) + b\tilde{X}_2(k)$
4.	Time shifting: $\tilde{x}(n + n_0)$ $\tilde{x}(n - n_0)$	$W_N^{kn_0}\tilde{X}(k)$ $W_N^{-kn_0}\tilde{X}(k)$

S. No.	Periodic Sequence (Period N)	DFS Coefficients
5.	Frequency shifting: $W_N^{k_0 n}\tilde{x}(n)$ $W_N^{-k_0 n}\tilde{x}(n)$	$\tilde{X}(k+k_0)$ $\tilde{X}(k-k_0)$
6.	Periodic convolution: $\sum_{m=0}^{N-1}\tilde{h}(m)\tilde{x}(n-m)$	$\tilde{H}(k)\tilde{X}(k)$
7.	Multiplication: $\tilde{x}_1(n)\tilde{x}_2(n)$	$\frac{1}{N}\sum_{m=0}^{N-1}\tilde{X}_1(k)X_2(k-m)$
8.	Symmetry Property: $\tilde{x}^*(n)$ $\tilde{x}^*(-n)$ $\text{Re}[\tilde{x}(n)]$ $I_m[\tilde{x}(n)]$ $\tilde{x}_{CS}(n)$ [conjugate symmetric part of $\tilde{x}(n)$] $\tilde{x}_{CAS}(n)$ [conjugate anti-symmetric part of $\tilde{x}(n)$]	$\tilde{X}^*(-k)$ $\tilde{X}^*(k)$ $\tilde{X}_{cs}(k)$ [conjugate symmetric part of $\tilde{X}(k)$] $\tilde{X}_{CAS}(k)$ [conjugate anti-symmetric part of $\tilde{X}(k)$] $\text{Re}[\tilde{X}(k)]$ $I_m[\tilde{X}(k)]$
9.	Symmetric property: If $\tilde{x}(n)$ is real	$\tilde{X}(k)=\tilde{X}(-k)$ $\text{Re}[\tilde{X}(k)]=\text{Re}[\tilde{X}(-k)]$ $I_m[\tilde{X}(k)]=-I_m[\tilde{X}(-k)]$ $\lvert\tilde{X}(k)\rvert=\lvert\tilde{X}(-k)\rvert$ $\underline{\tilde{X}(k)}=\underline{\tilde{X}(-k)}$

Table 2.1

Fourier Representation of Finite Duration Sequences: Discrete Fourier Transform

Let x(n) be a finite duration sequence. The N-point DFT of the sequence x(n) is represented by:

$$X(k)=\sum_{n=0}^{N-1}x(n)W_N^{kN} \quad , k=0,1,---N-1$$

Inverse Discrete Fourier Transform (IDFT) is:

$$x(n)=\frac{1}{N}\sum_{k=0}^{N-1}X(k)W_N^{-kN} \quad , n=0,1,---N-1$$

Properties of DFT

1. **Periodicity**
   If X(k) is N-point DFT of x(n), then
   x(n + N) = x(n) for all n
   X(k + N) = X(k) for all k
2. **Linearity**
   If, $x_1(n)\xleftrightarrow[N]{DFT}X_1(k)$
   $x_2(n)\xleftrightarrow[N]{DFT}X_2(k)$
   then for any real valued or complex valued constants a and b

$$ax_1(n) + bx_2(n) \xleftrightarrow[N]{DFT} aX_1(k) + bX_2(k)$$

3. Circular symmetries of a sequence

The N-point DFT of a finite duration sequences x(n) of length $l \le N$ is equivalent to the N-point DFT of a periodic sequence $x_p(n)$ of period N, which is obtained by periodically extending x(n), that is:

$$x_p(n) = \sum_{l=-\infty}^{\infty} x(n - lN)$$

Now, shift the periodic sequence $x_p(n)$ by k units to the right. Then, we can obtain another periodic sequence:

$$x'_p(n) = x_p(n-k) = \sum_{l=-\infty}^{\infty} x(n-k-ln)$$

The finite duration sequence:

$$x'(n) = \begin{cases} x'_p(n) & 0 \le n \le N-1 \\ 0 & \text{otherwise} \end{cases}$$

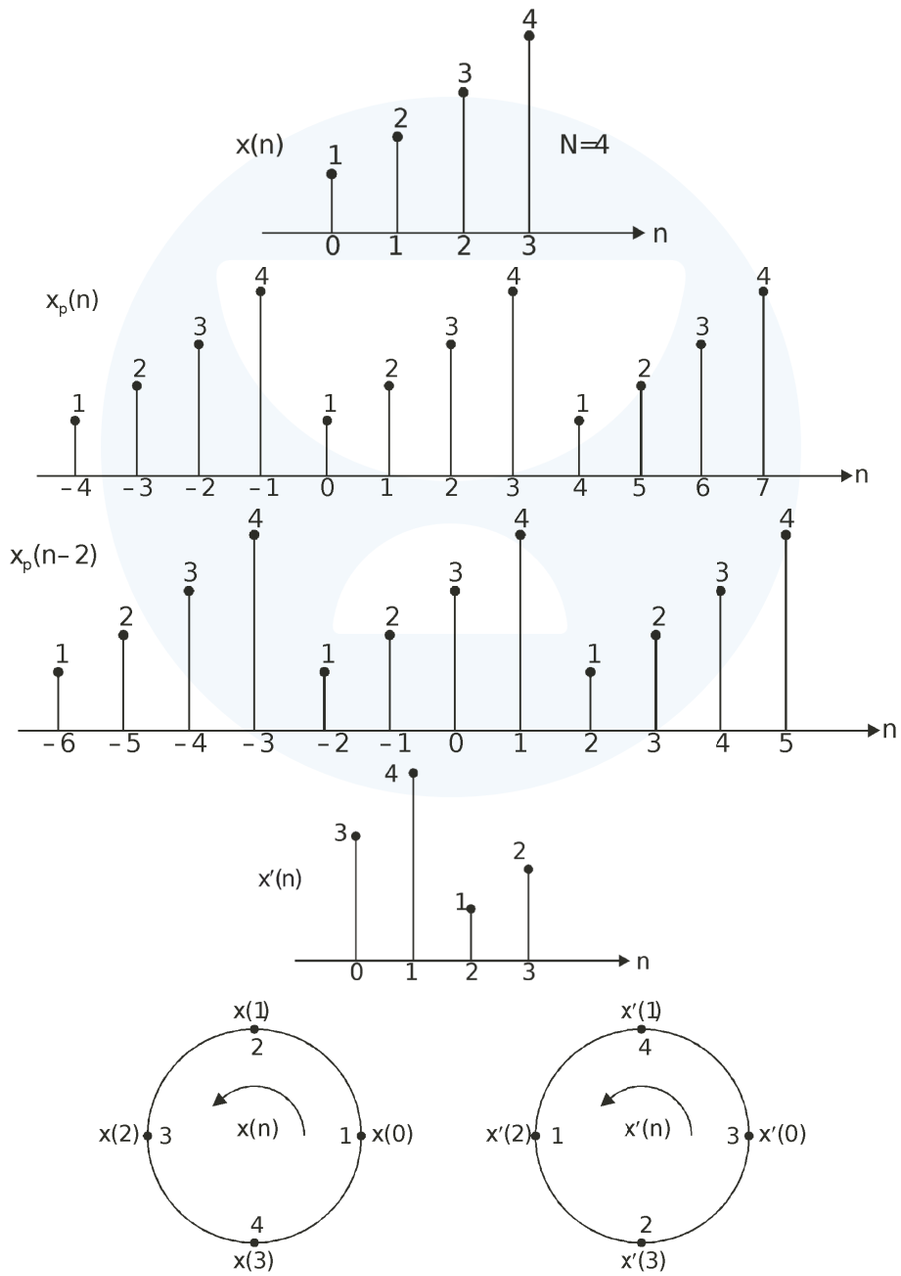


Fig. 2.1

In general, the circular shift of the sequence can be represented as the index module N. Thus, we may write:

$x'(n) = x(n - k, \text{ module } N)$

$\equiv x(n - k)N$

Example:

If $k = 2$

$x'(n) = x((n - 2))4$

$x'(0) = x((-2))4 = x(2)$

$x'(1) = x((-1))4 = x(3)$

$x'(2) = x((0))4 = x(0)$

$x'(3) = x((1))4 = x(1)$

Here, $x'(n)$ is simply $x(n)$ shifted circularly by two units in time, where the counter clockwise direction has been arbitrarily selected as positive direction.

- An N-point sequence is called circular even if it is symmetric about the point zero on the circle. This implies that:

  $x(N - n) = x(n), \quad 1 \le n \le N - 1$

- A complex sequence $x(n)$ is said to be conjugate symmetric if $x(n) = x^*(-n) = x^*(N - n)$

  An N-point sequence is called circularly odd, if it is anti-symmetric about the point zero on the circle. This implies that:

  $x(N - n) = -x(n) \quad 1 \le n \le N - 1$

- A complex sequence $x(n)$ is said to be conjugate anti-symmetric if $x(n) = x^*(-n) = -x^*(N - n)$
- The time reversal of an N-point sequence is attained by reversing its sample about the point zero on the circle.

  $x\left((-n)\right)_N = x(N - n) \quad 0 \le n \le N - 1$

  Note: The time reversal is equivalent to plotting $x(n)$ in a clockwise direction on a circle.

4. Symmetry properties of DFT

Let us assume that the N-point sequence $x(n)$ and its DFT are both complex valued, then the sequence can be expressed as:

$x(n) = x_R(n) + jx_I(n) \qquad 0 \le n \le N - 1$

$X(k) = X_R(k) + jX_I(k) \qquad 0 \le k \le N - 1$

$x_{ce}(n) =$

$\frac{1}{2}[x(n) + x((-n))_N] = \frac{1}{2}[x(n) + x(N - n)], \quad 0 \le n \le N - 1$

circularly even part of $x(n)$

$x_{ce}(n) =$

$\frac{1}{2}[x(n) - x((-n))_N] = \frac{1}{2}[x(n) - x(N - n)], \quad 0 \le n \le N - 1$

circularly odd part of $x(n)$

DFT

$$X(k) = \sum_{n=0}^{N-1} x(n)\, W_N^{kn}$$

$$= \sum_{n=0}^{N-1} x(n)\, e^{\frac{-j2\pi kn}{N}}$$

$$= \sum_{n=0}^{N-1} [x_R(n) + jx_I(n)] \left[\cos\left(\frac{2\pi kn}{N}\right) - j\sin\left(\frac{2\pi kn}{N}\right)\right]$$

$$= \sum_{n=0}^{N-1} \left[x_R(n)\cos\left(\frac{2\pi\, kn}{N}\right) + x_I(n)\sin\left(\frac{2\pi\, kn}{N}\right)\right] - j$$

$$\sum_{n=0}^{N-1} \left[x_R(n)\sin\left(\frac{2\pi\, kn}{N}\right) - x_I(n)\cos\left(\frac{2\pi\, kn}{N}\right)\right]$$

$X(k) = X_R(k) + jX_I(k) \qquad 0 \le k \le N - 1$

$$\boxed{\begin{aligned} X_R(k) &= \sum_{n=0}^{N-1} \left[x_R(n)\cos\left(\frac{2\pi kn}{N}\right) + x_I(n)\,\sin\left(\frac{2\pi kn}{N}\right)\right] \\ X_I(k) &= \sum_{n=0}^{N-1} \left[x_R(n)\sin\left(\frac{2\pi kn}{N}\right) - x_I(n)\,\cos\left(\frac{2\pi\, kn}{N}\right)\right] \end{aligned}}$$

IDFT

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\;\; W_N^{-kn} \qquad n = 0, 1, - - - N - 1$$

$$= \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, e^{j\left(\frac{2\pi kn}{N}\right)}$$

$$= \frac{1}{N}\sum_{k=0}^{N-1} [X_R(k) + jX_I(k)] \left[\cos\left(\frac{2\pi kn}{N}\right) + j\sin\left(\frac{2\pi kn}{N}\right)\right]$$

$$= \frac{1}{N}\sum_{k=0}^{N-1} \left[X_R(k)\cos\left(\frac{2\pi kn}{N}\right) - X_I(k)\,\sin\left(\frac{2\pi kn}{N}\right)\right] + j$$

$$\frac{1}{N}\sum_{k=0}^{N-1} \left[X_R(k)\sin\left(\frac{2\pi kn}{N}\right) + X_I(k)\,\cos\left(\frac{2\pi kn}{N}\right)\right]$$

$\because \quad x(n) = X_R(n) + jX_I(n) \quad 0 \le n \le N - 1$

$$X_R(n) = \frac{1}{N}\sum_{k=0}^{N-1}\left[X_R(k)\cos\left(\frac{2\pi kn}{N}\right) - X_I(k)\sin\left(\frac{2\pi kn}{N}\right)\right]$$

$$X_I(n) = \frac{1}{N}\sum_{k=0}^{N-1}\left[X_R(k)\sin\left(\frac{2\pi kn}{N}\right) + X_I(k)\cos\left(\frac{2\pi kn}{N}\right)\right]$$

Real Value Sequence

If x(n) is real, $X(N-k) = X^*(k) = X(-k)$

$|X(N-k)| = |X(k)|$

$\boxed{|X(N-k)} = -\boxed{|X(k)}$

$X_I(n) = 0$

$x(n) = x_R(n)$

$\boxed{x(n) = x_R(n)}$

Real and Even Sequence

If x(n) is real and even,

$x(n) = x(N-n) \qquad 0 \le n \le N-1$

$X_I(k) = 0$

$$X(k) = \sum_{n=0}^{N-1} x(n)\cos\left(\frac{2\pi kn}{N}\right), \quad 0 \le k \le N-1$$

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\cos\left(\frac{2\pi kn}{N}\right), \quad 0 \le n \le N-1$$

Real and Odd Sequence

If x(n) is real odd,

$x(n) = -x(N-n) \qquad 0 \le n \le N-1$

$XR(k) = 0$

$$X(k) = -j\sum_{n=0}^{N-1} x(n)\sin\left(\frac{2\pi kn}{N}\right) \quad 0 \le k \le N-1$$

$$x(n) = j\frac{1}{N}\sum_{k=0}^{N-1} X(k)\sin\left(\frac{2\pi kn}{N}\right) \quad 0 \le n \le N-1$$

Purely Imaginary Sequences

$x(n) = jx_I(n)$

$$X_R(k) = \sum_{n=0}^{N-1} x_I(n)\sin\left(\frac{2\pi kn}{N}\right)$$

$$X_I(k) = \sum_{n=0}^{N-1} X_I(n)\cos\left(\frac{2\pi kn}{N}\right)$$

If $X_I(n)$ is odd, then $X_I(k) = 0$, and hence, X(k) is purely real.

If $X_I(n)$ is even then $X_R(k) = 0$, and hence, X(k) is purely imaginary.

5. Circular convolution

$x_1(n)$ and $x_2(n)$ are two finite duration sequences of length N.

$$x_1(n) \xleftrightarrow[N]{DFT} X_1(k)$$

$$x_2(n) \xleftrightarrow[N]{DFT} X_2(k)$$

$$x_1(n) \textcircled{N} x_2(n) \xleftrightarrow[N]{DFT} X_1(k)\,X_2(k)$$

$$X_1(k) = \sum_{n=0}^{N-1} x_1(n)\,e^{\frac{-j2\pi kn}{N}} \qquad k = 0, 1, \dots\dots N-1$$

$$X_2(k) = \sum_{n=0}^{N-1} x_2(n)\,e^{\frac{-j2\pi kn}{N}}, \quad k = 0, 1, \dots\dots N-1$$

$$X_3(k) = X_1(k)\,X_2(k)$$

$$x_3(m) = \frac{1}{N}\sum_{k=0}^{N-1} X_3(k)\,e^{\frac{j2\pi km}{N}}$$

$$= \frac{1}{N}\sum_{k=0}^{N-1} x_1(k)\,X_2(k)\,e^{\frac{j2\pi km}{N}}$$

$$x_3(m) = \frac{1}{N}\sum_{k=0}^{N-1}\left[\sum_{n=0}^{N-1} x_1(n)e^{\frac{-j2\pi kn}{N}}\right]\left[\sum_{l=0}^{N-1} x_2(l)\,e^{\frac{-j2\pi kl}{N}}\right]e^{\frac{j2\pi km}{N}}$$

$$= \frac{1}{N}\sum_{n=0}^{N-1} x_1(n)\sum_{l=0}^{N-1} x_2(l)\left[\sum_{k=0}^{N-1} e^{j2\pi k(m-n-l)/N}\right]$$

$$\sum_{k=0}^{N-1} a^k = \begin{cases} N & a = 1 \\ \dfrac{1-a^N}{1-a} & a \ne 1 \end{cases}$$

if, $a = e^{j2\pi(m-n-l)/N}$

$m - n - l = pN \qquad \{p \text{ is an integer}\}$

$a = e^{j2\pi pN/N}$

$a = e^{j2p\pi} = 1$

$$\sum_{k=0}^{N-1} a^k = \begin{cases} N & l = m-n-PN = ((m-n))_N \\ 0 & \text{otherwise} \end{cases}$$

$$\boxed{x_3(m) = \sum_{n=0}^{N-1} x_1(n)x_2((m-n))_N} \qquad m = 0, 1, \dots N-1$$

6. Time reversal of a sequence

$$x(n) \xleftrightarrow[N]{DFT} X(k)$$

$$x((-n))_N = x(N-n) \xleftrightarrow[N]{DFT} X((-k))_N = X(N-k)$$

7. Circular time shift of a sequence

$$x(n) \xleftrightarrow[N]{DFT} X(k)$$

$$x((n-l))_N \xleftrightarrow[N]{DFT} e^{-j2\pi kl/N}X(k)$$

8. Circular frequency shift

$$x(n) \xleftrightarrow[N]{DFT} X(k)$$

$$e^{j2\pi ln/N}x(n) \xleftrightarrow[N]{DFT} X((k-l))_N$$

9. Complex conjugate property

$$x(n) \xleftrightarrow[N]{DFT} X(k)$$

$$x^*(n) \xleftrightarrow[N]{DFT} X^*((-k))_N = X^*(N-k)$$

10. Circular correlation

$$x(n) \xleftrightarrow[N]{DFT} X(k)$$

$$y(n) \xleftrightarrow[N]{DFT} Y(k)$$

$$\tilde{\gamma}_{xy}(l) \xleftrightarrow[N]{DFT} \tilde{R}_{xy}(k) = X(k)Y^*(k)$$

$\tilde{\gamma}_{xy}(l) \Rightarrow$ Unnormalised circular cross correlation sequence

11. Multiplication of two sequences

$$x_1(n) \xleftrightarrow[N]{DFT} X_1(k)$$

$$x_2(n) \xleftrightarrow[N]{DFT} X_2(k)$$

$$x_1(n)x_2(n) \xleftrightarrow[N]{DFT} \frac{1}{N} X_1(k) \textcircled{N} X_2(k)$$

12. Parseval's theorem

For complex valued sequences x(n) and y(n):

$$x(n) \xleftrightarrow[N]{DFT} X(k)$$

$$y(n) \xleftrightarrow[N]{DFT} Y(k)$$

$$\boxed{\sum_{n=0}^{N-1} x(n)y^*(n) = \frac{1}{N}\sum_{k=0}^{N-1} x(k)Y^*(k)}$$

$$x(n) = x_R^e(n) + x_R^0(n) + jx_I^e(n) + jx_I^0(n)$$

$$X(k) = X_R^e(k) + X_R^0(k) + jX_I^e(k) + jX_I^0(k)$$

Note:

- If x(n) is real and even, then X(k) is real and even.
- If x(n) is real and odd, then X(k) is imaginary and odd.
- If x(n) is imaginary and odd, then X(k) is real and odd.
- If x(n) is conjugate symmetric, then X(k) is real.
- If x(n) is conjugate anti-symmetric, then X(k) is imaginary.

13. Duality

$$x(n) \xleftrightarrow[N]{DFT} X(k)$$

$$X(n) \xleftrightarrow[N]{DFT} Nx((-k))_N$$

Summary of the Properties of DFT

Property	Time Domain	Frequency Domain
Notation	x(n), y(n)	X(k) , Y(k)
Periodicity	$x(n) = x(n+N)$	$X(k) = X(k+N)$
Linearity	$ax_1(n) + bx_2(n)$	$aX_1(k) + bX_2(k)$
The Reversal	$X((-n))_N = x(N-n)$	$X((-k)_N = X(N-k)$
Circular Time Shift	$X((n-l))_N$	$e^{-j2\pi kl/N}X(k)$
Circular Frequency shift	$e^{j2\pi ln/N}x(n)$	$X((k-l)_N$
Complex conjugate	$X^*(n)$	$X^*(N-k)$
Circular convolution	$X_1(n) \textcircled{N} x_2(n)$	$X_1(k)\ X_2(k)$
Multiplication of two sequences	$x_1(n)x_2(n)$	$\frac{1}{N}X_1(k)\ (N)\ X_2(k)$
Parseval's Theorem	$\sum_{n=0}^{N-1} x(n)y^*(n)$	$\frac{1}{N}\sum_{k=0}^{N-1} X(k)Y^*(k)$
Circular correlation	$x(n) \textcircled{N} y^*(-n)$	$X(k)Y^*(k)$
Duality	X(n)	$Nx((-k))_N$

Table 2.2

DFT as Linear Transformation

$$X(k) = \sum_{n=0}^{N-1} x(n)W_N^{kn} \quad , k = 0, 1---N-1$$

$$x(n) = \frac{1}{N}\sum_{n=0}^{N-1} x(k)W_N^{-kn} \quad , n = 0, 1, ---N-1$$

$W_N = e^{-j2\pi/N}$ is an Nth root of unity.

Note:

1. Each point of DFT can be accomplished by N complex multiplications and (N – 1) complex additions.
2. The N-point DFT values can be computed in a total of N^2 complex multiplications and N(N – 1) complex additions.
   Let us define an N-point vector xN of the signal sequence x(n), n = 0, 1, ... N – 1, an N-point vector XN of frequency samples and an NXN matrix WN as:

$$x_N = \begin{bmatrix} x(0) \\ x(1) \\ \vdots \\ x(N-1) \end{bmatrix}, \quad X_N = \begin{bmatrix} x(0) \\ X(1) \\ \vdots \\ x(N-1) \end{bmatrix}$$

$$W_N = \begin{bmatrix} 1 & 1 & 1 & --- & 1 \\ 1 & W_N^1 & W_N^2 & --- & W_N^{N-1} \\ 1 & W_N^2 & W_N^4 & --- & W_N^{2(N-1)} \\ \vdots & & & & \\ 1 & W_N^{N-1} & W_N^{2(N-1)} & --- & W_N^{(N-1)(N-1)} \end{bmatrix}$$

XN = WNxN

WN ⇒ Matrix of linear transformation

xN = WN–1 XN ... (i)

Inverse discrete Fourier transform can be expressed in matrix form as:

$$xN = \frac{1}{N} W_N^* X_n \quad \text{... (ii)}$$

From equation (i) and (ii)

$$WN{-}1 = \frac{1}{N} W_N^*$$

$$\boxed{W_N W_N^* = N I_N}$$

IN ⇒ NXN identity matrix

Solved Examples

2. Compute the DFT for the following four point sequence.

$x(n) = (\underset{\uparrow}{0}, 1, 2, 3)$

Solution:

$X_N = W_N x_N$

N = 4

$$X_4 = W_4 x_4$$

$$= \begin{bmatrix} W_4^0 & W_4^0 & W_4^0 & W_4^0 \\ W_4^0 & W_4^1 & W_4^2 & W_4^3 \\ W_4^0 & W_4^2 & W_4^4 & W_4^6 \\ W_4^0 & W_4^3 & W_4^6 & W_4^9 \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

$$= \begin{bmatrix} 1 & 1 & 1 & 1 \\ 1 & -j & -1 & j \\ 1 & -1 & 1 & -1 \\ 1 & j & -1 & -j \end{bmatrix} \begin{bmatrix} 0 \\ 1 \\ 2 \\ 3 \end{bmatrix}$$

$$X_4 = \begin{bmatrix} 6 \\ -2+2j \\ -2 \\ -2-2j \end{bmatrix}$$

3. Determine the DFT of the following sequence.

$$x[n] = \begin{cases} \frac{1}{4} & 0 \le n \le 2 \\ 0 & \text{otherwise} \end{cases}$$

Solution:

The N-point DFT of the sequence x(n) is defined as:

$$X(K) = \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi kn/N} \quad k = 0, 1, ----N-1$$

$$x(n) = \left\{ \underset{\uparrow}{\frac{1}{4}}, \frac{1}{4}, \frac{1}{4} \right\}$$

N = 3

$$x(k) = \sum_{n=0}^{2} x(n)\, e^{-j2\pi kn/N}$$

$$= \frac{1}{4}\left[1 + e^{\frac{-j2\pi k}{3}} + e^{\frac{-j4\pi k}{3}}\right]$$

$$= \frac{1}{4} e^{\frac{-j2\pi k}{3}} \left[e^{\frac{j2\pi k}{3}} + 1 + e^{\frac{-j2\pi k}{3}}\right]$$

$$\boxed{x(k) = \frac{1}{4} e^{\frac{-j2\pi k}{3}} \left[1 + 2\cos\left(\frac{2\pi k}{3}\right)\right]}, \; k = 0, 1, 2$$

4. Compute the circular convolution of the following two sequences.

$x_1(n) = \left\{ \underset{\uparrow}{2}, 1, 2, 1 \right\}$

$x_2(n) = \left\{ \underset{\uparrow}{1}, 2, 3, 4 \right\}$

Solution:

$x_3(m) = x_1(n)$ (N) $x_2(n)$

$$= \sum_{n=0}^{N-1} x_1(n)x_2((m-n))_N,\ m = 0, 1, ---N-1$$

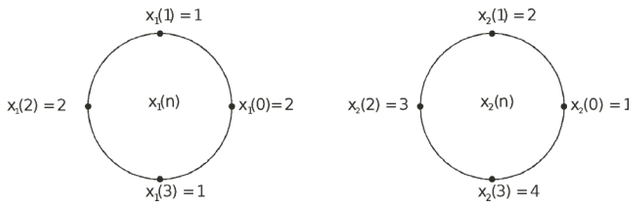


Fig. 2.2

$$x_3(m) = \sum_{m=0}^{3} x_1(n)x_2((m-n))_4$$

$$x_3(0) = \sum_{n=0}^{3} x_1(n)x_2((-n))_4$$

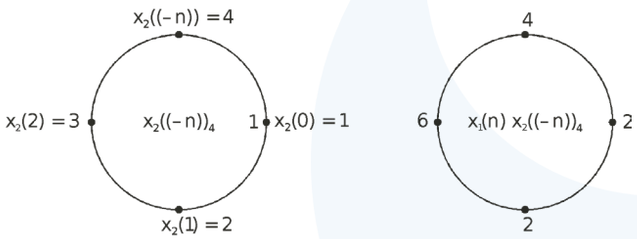


Fig. 2.3

$\boxed{x_3(o) = 14}$

$$x_3(1) = \sum_{n=0}^{3} x_1(n)\ x_2((1-n))_4$$

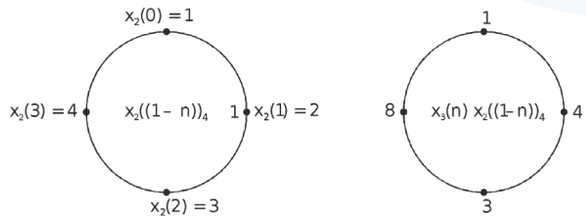


Fig. 2.4

$x_3(1) = 4 + 1 + 8 + 3$

$\boxed{x_3(1) = 16}$

$$x_3(2) = \sum_{n=0}^{3} x_1(n)x_2((2-n))_4$$

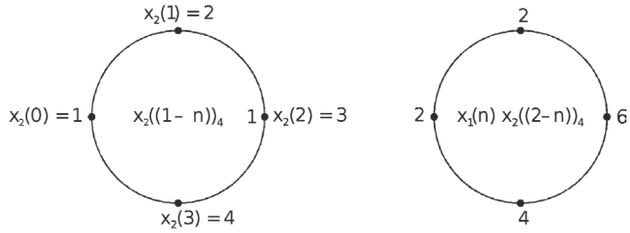


Fig. 2.5

$x_3(2) = 6 + 2 + 2 + 4$

$\boxed{x_3(2) = 14}$

$$x_3(3) = \sum_{n=0}^{3} x_1(n)\ x_2((3-n))_4$$

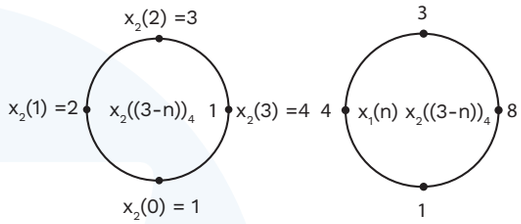


Fig. 2.6

$x_3(3) = 8 + 3 + 4 + 1$

$\boxed{x_3(3) = 16}$

$x_3(n) = \{14, 16, 14, 16\}$ (↑ at 14)

5. Compute the N-point DFT of each of the following sequences

a. $x_1(n) = \delta(n)$

b. $x_2(n) = \delta(n - n_0)$ $\quad 0 < n_0 < N$

c. $x_3(n) = \alpha^n$ $\quad 0 \le n < N$

d. $x_4(n) = u(n) - u(n - n_0)$ $\quad 0 < n_0 < N$

Solution:

$x_1(n) = \delta(n)$

a. $$X_1(k) = \sum_{n=0}^{N-1} \delta(n)\ W_N^{kn} \qquad k = 0, 1, ----N-1$$

$\boxed{X_1(k) = 1}$ $\qquad k = 0, 1, ---- N - 1$

b. $x_2(n) = \delta\ (n - n_0)$ $\qquad 0 < n_0 < N$

From circular time shift property

$x(n) \xleftrightarrow[N]{DFT} X(k)$

$$x((n-l))_N \xleftrightarrow[N]{DFT} e^{-j2\pi kl/N}X(k)$$

$$\delta(n) \xrightarrow[N]{DFT} 1$$

$$\delta(n-n_0) \xleftrightarrow[N]{DFT} e^{-j2\pi kn_0/N}$$

c. $x_3(n) = \alpha^n \quad 0 \le n < N$

$$X_3(k) = \sum_{n=0}^{N-1} x_3(n)\, e^{-j2\pi kn/N}$$

$$= \sum_{n=0}^{N-1} \alpha^n\, e^{-j2\pi kn/N}$$

$$X_3(k) = \sum_{n=0}^{N-1} \left(\alpha e^{-j2\pi k/N}\right)^n$$

$$= 1 + \left(\alpha e^{-j2\pi k/N}\right) + \left(\alpha e^{-j2\pi k/N}\right)^2 + ----N \text{ term}$$

$$X_3(k) = \frac{1-\left(\alpha e^{-j2\pi k/N}\right)^N}{1-\left(\alpha e^{-j2\pi k/N}\right)}$$

$$= \frac{1-\alpha^N \cdot e^{-j2\pi k}}{1-\left(\alpha e^{-j2\pi k/N}\right)}$$

$$\boxed{X_3(k) = \frac{1-\alpha^N}{1-\alpha e^{-j2\pi k/N}}}, \qquad k = 0, 1, ----- N-1$$

d. $x_4(n) = u(n) - u(n-n_0) \qquad 0 < n_0 < N$

$$X_4(k) = \sum_{n=0}^{n_0-1} e^{-j2\pi kn/N} = \sum_{n=0}^{N-1} \left(e^{\frac{-j2\pi k}{N}}\right)^n$$

$$= 1 + e^{-j2\pi k/N} + \left(e^{\frac{-j2\pi k}{N}}\right)^2 + \left(e^{\frac{-j2\pi k}{N}}\right)^3 + ---n_0 \text{ term}$$

$$= \frac{1-\left(e^{\frac{-j2\pi k}{N}}\right)^{n_0}}{1-e^{\frac{-j2\pi k}{N}}}$$

$$\boxed{x_4(k) = \frac{1-e^{\frac{-j2\pi kn_0}{N}}}{1-e^{\frac{-j2\pi k}{N}}}}$$

6. Determine IDFT of $X(k) = \{\underset{\uparrow}{3}, (2+j), 1, (2-j)\}$

Solution:

$N = 4$

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k) e^{j2\pi kn/N}$$

$$x(n) = \frac{1}{4}\sum_{k=0}^{3} X(k)\, e^{j2\pi\, kn/4}$$

$$x(0) = \frac{1}{4}\sum_{k=0}^{3} X(k) = \frac{1}{4}[3+2+j+1+2-j]$$

$$\boxed{x(0) = 2}$$

$$x(1) = \frac{1}{4}\sum_{k=0}^{3} X(k)\, e^{\frac{j2\pi k}{4}}$$

$$= \frac{1}{4}\sum_{k=0}^{3} X(k)\, e^{\frac{j\pi}{2}k}$$

$$= \frac{1}{4}\left[x(0) + x(1)e^{\frac{j\pi}{2}} + x(2)e^{j\pi} + x(3)e^{\frac{j3\pi}{2}}\right]$$

$$= \frac{1}{4}[3 + (2+j)j + 1(-1) + (2-j)(-j)]$$

$$\boxed{x(1) = 0}$$

$$x(2) = \frac{1}{4}\sum_{k=0}^{3} X(k)\, e^{j\pi k}$$

$$= \frac{1}{4}\left[X(0) + X(1)e^{j\pi} + X(2)e^{j2\pi} + X(3)e^{j3\pi}\right]$$

$$= \frac{1}{4}[3 + (2+j)(-1) + 1 + (2-j)(-1)]$$

$$\boxed{x(2) = 0}$$

$$x(3) = \frac{1}{4}\sum_{k=0}^{3} X(k)\, e^{j3\pi k/2}$$

$$= \frac{1}{4}\left[X(0) + X(1)e^{\frac{j3\pi}{2}} + X(2)e^{j3\pi} + X(3)e^{\frac{j9\pi}{2}}\right]$$

$$= \frac{1}{4}\left[3 + (2+j)(-j) - 1 + (2-j)(j)\right] = 1$$

$$\boxed{x(3) = 1}$$

$$x(n) = \left\{\underset{\uparrow}{2}, 0, 0, 1\right\}$$

7. Derive the DFT of the sample data sequence x(n) = {1, 1, 2, 2, 3, 3} and compute the corresponding amplitude and phase spectrum.

Solution:

$x(n) = \{1, 1, 2, 2, 3, 3\}$

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi kn/N} \qquad k = 0, 1, ---N-1$$

$N = 6$

$$X(k) = \sum_{n=0}^{5} x(n)e^{-j2\pi kn/6}$$

$$X(0) = \sum_{n=0}^{5} x(n) = 1+1+2+2+3+3$$

$X(0) = 12$

$|X(0)| = 12, \quad \angle X(0) = \tan^{-1}(0) = 0$

$$X(1) = \sum_{n=0}^{5} x(n)\, e^{\frac{-j2\pi n}{6}}$$

$$= x(0) + x(1)\, e^{\frac{-j\pi}{3}} + x(2)\, e^{\frac{-j2\pi}{3}} + x(3)e^{-j\pi} + x(4)e^{\frac{-j4\pi}{3}} + x(5)e^{\frac{-j5\pi}{3}}$$

$= 1 + (0.5 - j\ 0.866) + 2(-0.5 - j\ 0.866) + 2(-1) + 3(-0.5 + j\ 0.866) + 3(0.5 + j\ 0.866)$

$X(1) = -1.5 + j\ 2.598$

$$|x(1)| = \sqrt{(-1.5)^2 + (2.598)^2} = 3$$

$$\angle X(1) = \tan^{-1}\left(\frac{2.598}{-1.5}\right) = 180° - \tan^{-1}\left(\frac{2.598}{1.5}\right)$$

$\angle x(1) = 120°$

$$X(2) = \sum_{n=0}^{5} x(n)\, e^{\frac{-j2\pi n}{3}}$$

$$= x(0) + x(1)\, e^{\frac{-j2\pi}{3}} + x(2)e^{\frac{-j4\pi}{3}} + x(3)\, e^{\frac{-j6\pi}{3}} + x(4)e^{\frac{-j8\pi}{3}} + x(5)\, e^{\frac{-j10\pi}{3}}$$

$= 1 + (-0.5 - j\ 0.866) + 2\,(-0.5 + j\ 0.866) + 2(1) + 3(-0.5 - j\ 0.866) + 3\,(-0.5 + j\ 0.866)$

$X(2) = -1.5 + j\ 0.866$

$$|X(2)| = \sqrt{(-1.5)^2 + (0.866)^2} = 1.732$$

$$\angle X(2) = \tan^{-1}\left(\frac{0.866}{-1.5}\right)$$

$$= 180° - \tan^{-1}\left(\frac{0.866}{1.5}\right)$$

$\angle X(2) = 150°$

$$X(3) = \sum_{n=0}^{5} x(n)\, e^{-j\pi n}$$

$$= x(0) + x(1)\, e^{-j\pi} + x(2)e^{-j2\pi} + x(3)\, e^{-j3\pi} + x(4)\, e^{-j4\pi} + x(5)\, e^{-j5\pi}$$

$= 1 - 1 + 2 - 2 + 3 - 3 = 0$

$X(3) = 0$

$$X(4) = \sum_{n=0}^{5} x(n)\, e^{\frac{-j4\pi n}{3}}$$

$$= x(0) + x(1)e^{\frac{-j4\pi}{3}} + x(2)e^{\frac{-j8\pi}{3}} + x(3)\, e^{-j4\pi} + x(4)e^{\frac{-j16\pi}{3}} + x(5)\, e^{\frac{-j20\pi}{3}}$$

$= 1 + (-0.5 + j\ 0.866) + 2(-0.5 - j\ 0.866) + 2(1) + 3(-0.5 - j\ 0.866) + 3(-0.5 - j\ 0.866)$

$X(4) = -1.5 - j\ 0.866$

$$|X(4)| = \sqrt{(-1.5)^2 + (0.866)^2} = 1.732$$

$$\angle X(4) = \tan^{-1}\left(\frac{-0.866}{-1.5}\right) = 180° + \tan^{-1}\left(\frac{0.866}{1.5}\right)$$

$\angle X(4) = 210°$

$$X(5) = \sum_{n=0}^{5} x(n)e^{\frac{-j5\pi n}{3}}$$

$$= (0) + x(1)\, e^{\frac{-j5\pi}{3}} + x(2)e^{\frac{-j10\pi}{3}} + x(3)e^{-j5\pi} + x(4)e^{\frac{-j20\pi}{3}} + x(5)e^{\frac{-j25\pi}{3}}$$

$= 1 + (-0.5 + j\ 0.866) + 2(-0.5 + j\ 0.866) + 2(-1) + 3(-0.5 - j\ 0.866) + 3(0.5 - j\ 0.866)$

$X(5) = -1.5 - j\ 2.598$

$$|X(5)| = \sqrt{(-1.5)^2 + (-2.598)^2} = 3$$

$$\angle X(5) = \tan^{-1}\left(\frac{-2.598}{-1.5}\right) = 180° + \tan^{-1}\left(\frac{2.598}{1.5}\right)$$

$= 240°$

8. Consider a finite length sequence $x(n) = \delta(n) + 2\,\delta(n-5)$.

a. Find the 10-point discrete Fourier transform of x(n).

b. Find the sequence that has a discrete Fourier transform.

$$Y(k) = e^{j2k\left(\frac{2\pi}{10}\right)}X(k)$$

c. Find the 10-point sequence y(n) that has a discrete Fourier Transform.

$Y(k) = X(k)\, W(k)$

Where, X(k) is the 10-point DFT of x(n) and W(k) is the 10-point DFT of the sequence

$$w(n) = \begin{cases} 1 & 0 \le n \le 6 \\ 0 & \text{otherwise} \end{cases}$$

Solution:

a. $x(n) = \delta(n) + 2\delta(n-5)$

$$X(k) = 1 + 2\, e^{-j10\pi k/10}$$

$$= 1 + 2e^{-j\pi k}$$

$$\boxed{X(k) = 1 + 2(-1)^k}$$

b. $Y(k) = e^{2\pi\left(\frac{2\pi}{10}\right)} X(k)$

$y(n) = x((n+2))_{10}$

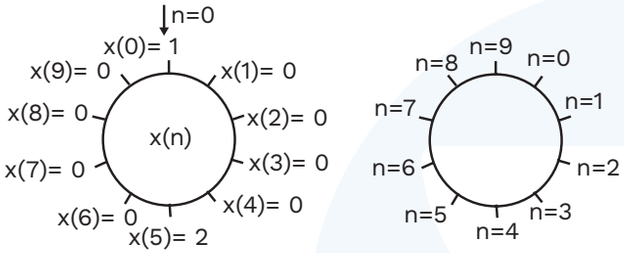


Fig. 2.7

$$x(n) = \left\{\underset{\uparrow}{1}, 0, 0, 0, 0, 2, 0, 0, 0, 0\right\}$$

$$x((n+2))_{10} = \left\{\underset{\uparrow}{0}, 0, 0, 2, 0, 0, 0, 0, 1, 0\right\}$$

$x(n) = \delta(n) + 2\delta(n-5)$

$x((n+2))_{10} = 2\delta(n-3) + \delta(n-8)$

c. $x_3(n) = x_1(n) \textcircled{N} x_2(n)$

$$x_3(m) = \sum_{n=0}^{N-1} x_1(n)\, x_2((m-n))_N$$

Using the matrix approach, we can formulate the circular convolution as:

$$\begin{bmatrix} x_1(0) & x_1(N-1) & x_1(N-2) & \dots & x_1(1) \\ x_1(1) & x_1(0) & x_1(N-2) & \dots & x_1(2) \\ \vdots & \vdots & \vdots & & \vdots \\ x_1(N-2) & x_1(N-3) & x_1(N-4) & \dots & x_1(N-1) \\ x_1(N-1) & x_1(N-2) & x_1(N-3) & \dots & x_1(0) \end{bmatrix} \begin{bmatrix} x_2(0) \\ x_2(1) \\ \vdots \\ x_2(N-2) \\ x_2(N-1) \end{bmatrix} = \begin{bmatrix} x_3(0) \\ x_3(1) \\ \vdots \\ x_3(N-2) \\ x_3(N-1) \end{bmatrix}$$

Fig. 2.8

$Y(k) = X(k)\, W(k)$

$y(n) = x(n) \textcircled{N} w(n)$

$$y(m) = \sum_{n=0}^{N-1} x(n)\, w((m-n))_N$$

$$\begin{bmatrix} x(0) & x(9) & x(8) & x(7) & x(6) & x(5) & x(4) & x(3) & x(2) & x(1) \\ x(1) & x(0) & x(9) & x(8) & x(7) & x(6) & x(5) & x(4) & x(3) & x(2) \\ x(2) & x(1) & x(0) & x(9) & x(8) & x(7) & x(6) & x(5) & x(4) & x(3) \\ x(3) & x(2) & x(1) & x(0) & x(9) & x(8) & x(7) & x(6) & x(5) & x(4) \\ x(4) & x(3) & x(2) & x(1) & x(0) & x(9) & x(8) & x(7) & x(6) & x(5) \\ x(5) & x(4) & x(3) & x(2) & x(1) & x(0) & x(9) & x(8) & x(7) & x(6) \\ x(6) & x(5) & x(4) & x(3) & x(2) & x(1) & x(0) & x(9) & x(8) & x(7) \\ x(7) & x(6) & x(5) & x(4) & x(3) & x(2) & x(1) & x(0) & x(9) & x(8) \\ x(8) & x(7) & x(6) & x(5) & x(4) & x(3) & x(2) & x(1) & x(0) & x(9) \\ x(9) & x(8) & x(7) & x(6) & x(5) & x(4) & x(3) & x(2) & x(1) & x(0) \end{bmatrix} \begin{bmatrix} w(0) \\ w(1) \\ w(2) \\ w(3) \\ w(4) \\ w(5) \\ w(6) \\ w(7) \\ w(8) \\ w(9) \end{bmatrix} = \begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \\ y(4) \\ y(5) \\ y(6) \\ y(7) \\ y(8) \\ y(9) \end{bmatrix}$$

$$\begin{bmatrix} 1 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 \\ 0 & 1 & 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 \\ 0 & 0 & 1 & 0 & 0 & 0 & 0 & 2 & 0 & 0 \\ 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 2 & 0 \\ 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 & 2 \\ 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 & 0 \\ 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 & 0 \\ 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 2 & 0 & 0 & 0 & 1 & 0 & 0 \\ 0 & 0 & 0 & 0 & 2 & 0 & 0 & 0 & 0 & 1 \end{bmatrix} \begin{bmatrix} 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 1 \\ 0 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \\ y(4) \\ y(5) \\ y(6) \\ y(7) \\ y(8) \\ y(9) \end{bmatrix} \begin{bmatrix} 3 \\ 3 \\ 1 \\ 1 \\ 1 \\ 3 \\ 3 \\ 2 \\ 2 \\ 2 \end{bmatrix} = \begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \\ y(4) \\ y(5) \\ y(6) \\ y(7) \\ y(8) \\ y(9) \end{bmatrix}$$

$$y(n) = \left\{\underset{\uparrow}{3}, 3, 1, 1, 1, 3, 3, 2, 2, 2\right\}$$

9. Compute the N-point DFT of $x(n) = \begin{cases} 1, & \text{for n even} \\ 0, & \text{for n odd} \end{cases}$

Solution:

$$X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j2\pi kn/N}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x(2n)\, e^{-j2\pi k\, 2n/N} + \sum_{n=0}^{\frac{N}{2}-1} x(2n+1)\, e^{-2\pi k(2n+1)/N}$$

$$= \sum_{n=0}^{\frac{N}{2}-1} x(2n)\, e^{-j4\pi kn/N} = \sum_{n=0}^{\frac{N}{2}-1} e^{-j4\pi kn/N}$$

$$X(k) = \frac{1-\left(e^{\frac{-j4\pi k}{N}}\right)^{\frac{N}{2}}}{1-e^{\frac{-j4\pi k}{N}}}$$

$$X(k) = \frac{1-e^{-j2\pi k}}{1-e^{\frac{-j4\pi k}{N}}}$$

10. Let x(n) be the sequence:

$x(n) = 2\,\delta(n) + (n-1) + \delta(n-3)$

The five-point DFT of x(n) is computed and resulting sequence is squared.

$Y(k) = X^2(k)$

A five-point DFT is then computed to produce the sequence y(n). Find the sequence y(n).

Solution:

$Y(k) = X^2(k)$

$y(n) = x(n) \textcircled{N} x(n)$

$$x(n) = \left\{\underset{\uparrow}{2}, 1, 0, 1, 0\right\}$$

$$\begin{bmatrix} x(0) & x(4) & x(3) & x(2) & x(1) \\ x(1) & x(0) & x(4) & x(3) & x(2) \\ x(2) & x(1) & x(0) & x(4) & x(3) \\ x(3) & x(2) & x(1) & x(0) & x(4) \\ x(4) & x(3) & x(2) & x(1) & x(0) \end{bmatrix} \begin{bmatrix} x(0) \\ x(1) \\ x(2) \\ x(3) \\ x(4) \end{bmatrix} = \begin{bmatrix} y(0) \\ y(1)` \\ y(2) \\ y(3) \\ y(4) \end{bmatrix}$$

$$\begin{bmatrix} 2 & 0 & 1 & 0 & 1 \\ 1 & 2 & 0 & 1 & 0 \\ 0 & 1 & 2 & 0 & 1 \\ 1 & 0 & 1 & 2 & 0 \\ 0 & 1 & 0 & 1 & 2 \end{bmatrix} \begin{bmatrix} 2 \\ 1 \\ 0 \\ 1 \\ 0 \end{bmatrix} = \begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \\ y(4) \end{bmatrix}$$

$$\begin{bmatrix} 4 \\ 5 \\ 1 \\ 4 \\ 2 \end{bmatrix} = \begin{bmatrix} y(0) \\ y(1) \\ y(2) \\ y(3) \\ y(4) \end{bmatrix}$$

$$y(n) = \left\{\underset{\uparrow}{4}, 5, 1, 4, 2\right\}$$

Twiddle Factor (WN) and Its Properties

$$W_N = e^{\frac{-j2\pi}{N}} = \cos\frac{2\pi}{N} - j\sin\frac{2\pi}{N}$$

$$\boxed{|W_N| = 1}$$

$$\boxed{|\underline{W_N} = -\frac{2\pi}{n}}$$

Property	Discription
1	$W_N^{N/4} = -j$
2	$W_N^{N/2} = -1$
3	$W_N^{3N/4} = j$
4	$W_N^{N} = 1$
5	$W_N^{K+N} = W_N^K \Rightarrow$ Periodicity property
6	$W_N^{K+N/2} = -W_N^K \Rightarrow$ Symmetry property
7	$W_N^{2K} = W_{N/2}^K$
8	$W_N^{*} = W_N^{-1}$

Table 2.3

Some Important Results

let X(k) denote the N-point DFT of x(n)

1. $\boxed{X(0) = \sum_{n=0}^{N-1} x(n)}$

Proof: $X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\left(\frac{2\pi}{N}\right)kn}$

$$\boxed{X(0) = \sum_{n=0}^{N-1} x(n)}$$

2. $X\left(\frac{N}{2}\right) = \sum_{n=0}^{N-1} (-1)^n x(n)$

Proof: $X(k) = \sum_{n=0}^{N-1} x(n)\, e^{-j\left(\frac{2\pi}{N}\right)kn}$

$$X(k)\big|_{k=N/2} = \sum_{n=0}^{N-1} x(n)\, e^{-j\left(\frac{2\pi}{N}\right)\cdot\frac{N}{2}\cdot n}$$

$$X\left(\frac{N}{2}\right) = \sum_{n=0}^{N-1} x(n)\, e^{-j\pi n}$$

$$\boxed{X\left(\frac{N}{2}\right) = \sum_{n=0}^{N-1} (-1)^n\, x(n)}$$

3. $\sum_{k=0}^{N-1} X(k) = Nx(0)$

Proof: $x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, e^{j\left(\frac{2\pi}{N}\right)kn}$

$$x(0) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)$$

$$\boxed{\sum_{k=0}^{N-1} x(k) = Nx(0)}$$

4. $\sum_{k=0}^{N-1} |X(k)|^2 = N\sum_{n=0}^{N-1} |x(n)|^2$

Proof: Using Parseval's Relation

$$\sum_{n=0}^{N-1} x(n)y^*(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, Y^*(k)$$

If $y(n) = x(n)$

$$\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N}\sum_{k=0}^{N-1} |X(k)|^2$$

$$\boxed{\sum_{k=0}^{N-1} |X(k)|^2 = N\sum_{n=0}^{N-1} |x(n)|^2}$$

Solved Examples

11. Consider a length – 12 sequence define for $0 \le n \le 11$

$$x(n) = \left\{\underset{\uparrow}{3}, -1, 2, 4, -3, -2, 0, 1, -4, 6, 2, 5\right\}$$

with a 12-point DFT given by X(k), $0 \le k \le 11$. Evaluate the following function of X(k) without computing the DFT

a. X(0)

b. X(6)

c. $\sum_{k=0}^{11} X(k)$

d. $\sum_{k=0}^{11} e^{-j4\pi k/6} X(k)$

e. $\sum_{k=0}^{11} |X(k)|^2$

Solution:

N = 12

a. $X(0) = \sum_{n=0}^{N-1} x(n) = \sum_{n=0}^{11} x(n)$

$= 3 - 1 + 2 + 4 - 3 - 2 + 0 + 1 - 4 + 6 + 2 + 5$

$X(0) = 13$

b. $X\left(\frac{N}{2}\right) = X(6) = \sum_{n=0}^{11} (-1)^n x(n)$

$= x(0) - x(1) + x(2) - x(3) + x(4) - x(5) + x(6) - x(7) + x(8) - x(9) + x(10) - x(11)$

$= 3 + 1 + 2 - 4 - 3 + 2 + 0 - 1 - 4 - 6 + 2 - 5$

$\boxed{X(6) = -13}$

c. $\sum_{k=0}^{N-1} X(k) = Nx(0)$

$\sum_{k=0}^{11} X(k) = 12\, x(0) = 12 \times 3 = 36$

$$\boxed{\sum_{k=0}^{11} X(k) = 36}$$

d. IDFT

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k)\, e^{j\left(\frac{2\pi}{N}\right)kn}$$

$$N\, x(n) = \sum_{k=0}^{N-1} e^{j\left(\frac{2\pi}{N}\right)kn} X(k)$$

$$12\, x(n) = \sum_{k=0}^{11} e^{j\left(\frac{2\pi}{12}\right)kn} X(k)$$

$$12\, x(n) = \sum_{k=0}^{11} e^{j\left(\frac{\pi}{6}\right)kn} X(k)$$

At n = –4 we get

$$12\, x((-4))_{12} = \sum_{k=0}^{11} e^{-j\left(\frac{4\pi}{6}\right)k} X(k)$$

$$\sum_{k=0}^{11} e^{\frac{-j4\pi}{6}k} X(k) = 12x((-4))_{12}$$

$$= 12\, x(12 - 4)$$

$$= 12\, x(8)$$

$$= 12\, (-4) = -48$$

$$\sum_{k=0}^{11} e^{-j\left(\frac{4\pi}{6}\right)k} X(k) = -48$$

e. $$\sum_{k=0}^{N-1} |X(k)|^2 = N\sum_{n=0}^{N-1} |x(n)|^2$$

$$\sum_{k=0}^{11} |X(k)|^2 = 12\sum_{n=0}^{11} |x(n)|^2$$

$$= 12[9 + 1 + 4 + 16 + 9 + 4 + 0 + 1 + 16 + 36 + 4 + 25]$$

$$\boxed{\sum_{k=0}^{11} |X(k)|^2 = 1500}$$

12. Let X(k) be a 14-point DFT of a length 14 real sequence x(n). The first eight samples are given by:

X(0) = 12, X(1) = –1 + j3, X(2) = 3 + j4, X(3) = 1 – j5

X(4) = –2 + j2, X(5) = 6 + j3, X(6) = –2 – j3, X(7) = 10

Determine the remaining samples of X(k). Evaluate the following function of x(n) without computing the IDFT of X(k).

a. X(0)

b. X(7)

c. $$\sum_{n=0}^{13} x(n)$$

d. $$\sum_{n=0}^{13} e^{\frac{j4\pi n}{7}} x(n)$$

e. $$\sum_{n=0}^{13} |x(n)|^2$$

Solution:

Since x(n) is real sequence

$X(k) = X^*((-k))_N = x^*(N - k)$

$X(8) = X^*((-8))_{14} = X^*(14 - 8) = X^*(6) = -2 + j3$

$X(9) = X^*((-9))_{14} = X^*(14 - 9) = X^*(5) = 6 - j3$

$X(10) = X^*((-10))_{14} = X^*(14 - 10) = X^*(4) = -2 - j2$

$X(11) = X^*((-11)_{14} = X^*(14 - 11) = X^*(3) = 1 + j5$

$X(12) = X^*((-12)_{14} = X^*(14 - 12) = X^*(2) = 3 - j4$

$X(13) = X^*((-13))_{14} = X^*(14 - 13) = X^*(1) = -1 - j3$

a. $$x(0) = \frac{1}{N}\sum_{n=0}^{N-1} X(k) = \frac{1}{14}\sum_{k=0}^{13} X(k) = \frac{32}{14} = \frac{16}{7}$$

$$\boxed{x(0) = \frac{16}{7}}$$

b. $$x\left(\frac{N}{2}\right) = x(7) = \frac{1}{N}\sum_{k=0}^{13} (-1)^k X(k)$$

$$= \frac{1}{14}\,[X(0) - X(1) + X(2) - X(3) + X(4) - X(5) + X(6) - X(7) + X(8) - X(9) + X(10) - X(11) + X(12) - X(13)]$$

$$= \frac{-12}{14} = \frac{-6}{7}$$

$$\boxed{x(7) = \frac{-6}{7}}$$

c. $$\sum_{n=0}^{N-1} x(n) = X(0)$$

$$\boxed{\sum_{n=0}^{13} x(n) = 12}$$

d. DFT

$$X(k) = \sum_{n=0}^{N-1} x(n) e^{-j\left(\frac{2\pi}{N}\right)kn}$$

$$= \sum_{n=0}^{13} x(n)\, e^{-j\left(\frac{2\pi}{14}\right)kn}$$

$$X(k) = \sum_{n=0}^{13} x(n)\, e^{-j\frac{\pi}{7}kn}$$

At k = −4,

$$\sum_{n=0}^{13} e^{j\frac{4\pi n}{7}} x(n) = X((-4))_{14} = X(14 - 4) = X(10)$$

$$\boxed{\sum_{n=0}^{13} e^{j\frac{4\pi n}{7}} x(n) = -2 - j2}$$

e. $$\sum_{n=0}^{N-1} |x(n)|^2 = \frac{1}{N}\sum_{k=0}^{N-1} |X(k)|^2$$

$$\sum_{n=0}^{13} |x(n)|^2 = \frac{1}{14}\sum_{k=0}^{13} |X(k)|^2 = \frac{498}{14}$$

$$\boxed{\sum_{n=0}^{13} |x(n)|^2 = \frac{498}{14}}$$

Comparison between Linear and Circular Convolution

S. No.	Linear Convolution	Circular Convolution
1.	$y(n) = x(n) * h(n)$ Length of $y(n) = L_1 + L_2 - 1$ $L_1 \rightarrow$ Length of x(n) $L_2 \rightarrow$ Length of h(n)	$y(n) = x(n) \textcircled{N} h(n)$ Length of $y(n) = \max(L_1, L_2)$ $L_1 \rightarrow$ Length of x(n) $L_2 \rightarrow$ Length of h(n)
2.	Lengths of x(n) and h(n) need not be same	Lengths of x(n) & h(n) must be same.
3.	Zero padding is not required	Zero padding is required if the length of the sequences are different.
4.	Linear time shift is used to perform the linear convolution	Circular time shift is used to perform the circular convolution.

Table 2.4

Solved Examples

13. If x(n) has an N point DFT X(k), find the N-point of:

$$y(n) = \cos\left(\frac{2\pi}{N}n\right)x(n)$$

Solution:

$$y(n) = \cos\left(\frac{2\pi}{N}n\right)x(n)$$

$$y(n) = \frac{1}{2}e^{j\left(\frac{2\pi}{N}\right)n}x(n) + \frac{1}{2}e^{-j\left(\frac{2\pi}{N}\right)n}x(n)$$

$$\boxed{Y(k) = \frac{1}{2}\left[x((k-1))_N + x((k+1))_N\right]}$$

14. Determine the N-point IDFT of $X(k) = \delta(k) = s(k)$, $0 \le k \le N-1$ and hence for $0 \le k_0 \le N-1$, determine the DFT of

a. $x_1(n) = e^{\frac{j2\pi k_0 n}{N}}$

b. $x_2(n) = e^{\frac{-j2\pi k_0 n}{N}}$

c. $x_3(n) = \cos\left(\frac{2\pi}{N}k_0 n\right)$

d. $x_4(n) = \sin\left(\frac{2\pi}{N}k_0 n\right)$

Solution:

$$x(n) = \frac{1}{N}\sum_{k=0}^{N-1} X(k) e^{j\left(\frac{2\pi}{N}\right)kn}$$

$$= \frac{1}{N}\sum_{k=0}^{N-1}\delta(k)e^{j\left(\frac{2\pi}{N}\right)kn} = \frac{1}{N}$$

$$\frac{1}{N} \xleftrightarrow[N]{DFT} \delta(k)$$

$$1 \xleftrightarrow[N]{DFT} N\delta(k)$$

a. $x_1(n) = e^{j\frac{2\pi}{N}k_0n}$

Using circular frequency shift property

$$e^{j2\pi ln/N}x(n) \xleftrightarrow[N]{DFT} X((k-l))_N$$

If $x(n) = 1$

$X(k) = N\,\delta\,(k)$

$$\boxed{e^{j\frac{2\pi}{N}k_0n} \xleftrightarrow[N]{DFT} N\delta((k-k_0))_N}$$

b. $x_2(n) = e^{\frac{-j2\pi k_0n}{N}}$

$$e^{-j\frac{2\pi}{N}k_0n} \xleftrightarrow[N]{DFT} N\delta((k+k_0))_N$$

c. $x_3(n) = \cos\left(\frac{2\pi}{N}k_0n\right)$

$$= \frac{1}{2}\left[e^{\frac{j2\pi}{N}k_0n} + e^{\frac{-j2\pi}{N}k_0n}\right]$$

$$\boxed{X_3(k) = \frac{N}{2}\left[\delta((k-k_0))_N + \delta((k+k_0))_N\right]}$$

d. $x_4(n) = \sin\left(\frac{2\pi}{N}k_0n\right)$

$$= \frac{1}{2j}\left[e^{\frac{j2\pi}{N}k_0n} - e^{\frac{-j2\pi}{N}k_0n}\right]$$

$$\boxed{x_4(k) = \frac{N}{2j}\left[\delta((k-k_0))_N - \delta((k+k_0))_N\right]}$$

15. Let $x(n) = \left\{\underset{\uparrow}{1}, 2, 0, 3\right\}$. Find the circularly folded signal $x_1(n) = x((-n))_4$, and also determine the circularly even and odd part of x(n).

Solution:

$$x(n) = \left\{\underset{\uparrow}{1}, 2, 0, 3\right\}$$

$N = 4$

$x_1(n) = x((-n))_4$

$x_1(n) = x(N - n)_4 = x(4 - n)_4$

$x_1(1) = x(3)_4 = 3$

$x_1(2) = x(2)_4 = 0$

$x_1(3) = x(1)_4 = 2$

$x((-n))_4 = \left\{\underset{\uparrow}{1}, 3, 0, 2\right\}$

Circularly Even Point of x(n):

$$x_{ce}(n) = \frac{1}{2}\left[x(n) + x((-n))_4\right]$$

$$x_{ce}(n) = \left\{\underset{\uparrow}{1}, 2.5, 0, 2.5\right\}$$

Circularly odd part of x(n):

$$x(n) = \frac{1}{2}\left[x(n) - x((-n))_4\right]$$

$$= \left\{\underset{\uparrow}{0}, -0.5, 0, 0.5\right\}$$

16. If $x(n) = \left\{\underset{\uparrow}{3}, 2, A, 0, B\right\}$ is circularly even, find A and B.

Solution:

x(n) is circularly even, i.e.,

$x(n) = x((-n))_5 = x(5 - n)_5$, $1 \le n \le 4$

$x(n) = x_1(n), 1 \le n \le 4$

where, $x_1(n) = x(5 - n)_5$

$x_1(1) = x(4)_5 = B$

$x_1(2) = x(3)_5 = 0$

$x_1(3) = x(2)_5 = A$

$x_1(4) = x(1)_5 = 2$

$x_1(n) = \left\{\underset{\uparrow}{3}, B, 0, A, 2\right\}$

$x(n) = \{3, 2, A, 0, B\}$

$x(n) = x_1(n)$, therefore,

$\boxed{A = 0, B = 2}$

17. Given the sequence x(n) represented by $x(n) = \left\{\underset{\uparrow}{1}, 2, 3, 4, 0\right\}$

Find the sequence y(n) whose length - 6 DFT is given by $Y(k) = Re\{X(k)\}$.

Solution:

$Y(k) = Re\{X(k)\}$

If x(n) is conjugate symmetry, then X(k) is real.

$$y(n) = \frac{1}{2}\left[x(n) + x^*((-n))_6\right]$$

$$y(n) = \frac{1}{2}\left[x(n) + x(6-n)\right]$$

$y(0) = 1$

$$y(1) = \frac{1}{2}[x(1) + x(5)] = \frac{1}{2}[2 + 0] = 1$$

$$y(2) = \frac{1}{2}[x(2) + x(4)] = \frac{1}{2}[3 + 0] = 1.5$$

$$y(3) = \frac{1}{2}[x(3) + x(3)] = \frac{1}{2}[4 + 4] = 4$$

$$y(4) = \frac{1}{2}[x(4) + x(2)] = \frac{1}{2}[0 + 3] = 1.5$$

$$y(5) = \frac{1}{2}[x(5) + x(1)] = \frac{1}{2}[0 + 2] = 1$$

$$y(n) = \left\{1, 1, 1.5, 4, 1.5, 1\right\}$$

18. Let X(k) denote the N-point DFT of an N-point sequence x(n). We represented the DFT operation as X(k) = F[x(n)]. Determine the sequence y(n) obtained by applying the DFT operation 6 times to x(n), i.e.,
y(n) = F[F[F[F[F[F[x(n)]]]]]]

Solution:

Using duality property:

$F[F[x(n)]] = Nx((-n))_N = x_1(n)$

$$F[F[\underbrace{F[F[x(n)]]}_{x_1(n)}]]$$

$$= F[F[x_1(n)]] = Nx_1((-n))_N$$

$$= N^2x(n) = x_2(n)$$

$$y(n) = \underbrace{F[F[F[F[F[x(n)]]]]]}_{x_2(n)}$$

$$= F[F[x_2(n)]]$$

$$= Nx_2((-n))_N = N^3x((-n))_N$$

Practice Exercise

Level I

1. The convolution of two signals in time domain is equivalent to _____ of their spectra in frequency domain.
(A) convolution
(B) multiplication
(C) summation
(D) division

2. In an N-point DFT of finite duration signal x(n) of length L, the value of N should be:
(A) $N \geq L$
(B) $N < L$
(C) $N = 0$
(D) N = L2

3. In an LTI system, if the input has N_x samples and impulse response has N_h samples, then the output will have N_y samples, where:
(A) $N_y = N_x - N_h$
(B) $N_y = N_x + N_h$
(C) $N_y = N_x + N_h + 1$
(D) $N_y = N_x + N_h - 1$

4. Circular convolution is also called:
(A) linear convolution
(B) periodic convolution
(C) nonlinear convolution
(D) analytical convolution

5. Appending zeros to a sequence to increase its length is called:
(A) circular convolution
(B) DFT
(C) zero padding
(D) windowing

6. In overlap-add method of sectioned convolution, the overlapped samples of the output are:
(A) added
(B) subtracted
(C) multiplied
(D) None of the above

7. In overlap-save method of sectioned convolution, the overlapped samples of one of the output sequences are:
(A) added

(B) subtracted
(C) discarded
(D) None of the above

8. The DFT supports:
(A) linear convolution
(B) circular convolution
(C) Both (A) and (B)
(D) None of the above

9. The algorithm used to compute any set of equally spaced samples of Fourier transform on the unit circle is:
(A) DFT algorithm
(B) FFT algorithm
(C) Goertzel algorithm
(D) chirp transform algorithm

10. The Fourier transform of a discrete-time periodic signal is:
(A) continuous and periodic, or non-periodic
(B) continuous and periodic
(C) discrete and non-periodic
(D) discrete and periodic

11. The N-point DFT of finite length signal $x(n) = \delta(n)$ is:
(A) 0
(B) z^{-1}
(C) 1
(D) W_N

12. The multiplication of two DFTs is same as:
(A) linear convolution of the sequences
(B) correlation of the sequences
(C) cross-correlation of the sequences
(D) circular convolution of the sequences

13. The wideband spectrogram results from a window that is short in time and characterised by:
(A) poor resolution in frequency dimension and good resolution in time dimension
(B) good resolution in frequency dimension and poor resolution in time dimension
(C) poor resolution in frequency dimension and poor resolution in time dimension
(D) good resolution in frequency dimension and good resolution in time dimension

14. If X(k) is the N-point DFT of length-N sequence x(n), then what is the N-point DFT of the sequence y(n) = X(n)?
(A) $x((-n))_N$
(B) $N\, x((-n))_N$
(C) Nx(n)
(D) $x^2(n)$

15. The N-point DFT of the sequence $x(n) = (-1)^n$, $0 \le n \le N-1$ (N is even) is:
(A) $\delta((-k))_N$
(B) $(-1)k\, X(k)$
(C) $N\delta\left(\left(k - \frac{N}{2}\right)\right)_N$
(D) $\delta\left(\left(k - \frac{N}{2}\right)\right)_N$

16. The N-point DFT of a finite duration sequence can be obtained as:
(A) $X(k) = X(z)\big|_{z=e^{j\frac{2\pi n}{N}}}$
(B) $X(k) = X(z)\big|_{z=e^{j\frac{2\pi kn}{N}}}$
(C) $X(k) = X(z)\big|_{z=e^{-j\frac{2\pi kn}{N}}}$
(D) $X(k) = X(z)\big|_{z=e^{j\frac{2\pi k}{N}}}$

17. Let X(k) be an N-point DFT of the length-N signal x(n) with sampling frequency f_s. The relationship between frequency bin and the frequency is given by:
(A) $f = \frac{kf_s}{N}$
(B) $f = \frac{kN}{f_s}$
(C) $f = \frac{k}{N}$
(D) $f = \frac{k}{f_s}$

Level II

1. The Discrete Fourier Transform (DFT) of the 4-point sequence
$x(n) = \{x[0], x[1], x[2], x[3]\} = \{3, 2, 3, 4\}$ is $X[k] = \{X[0], X[1], X[2], X[3]\} = \{12, 2j, 0, -2j\}$. If $X_1[k]$ is the DFT of the 12-point sequence $x_1[n] = \{3, 0, 0, 2, 0, 0, 3, 0, 0, 4, 0, 0\}$, the value of $\left|\frac{X_1[8]}{X_1[11]}\right|$ is _____.

2. Consider two real sequences with time-origin marked by the bold value.
$x_1[n] = \{1, 2, 3, 0\}$, $x_2[n] = \{1, 3, 2, 1\}$
Let $X_1(k)$ and $X_2(k)$ be 4-point DFT's of $x_1[n]$ and $x_2[n]$ respectively.
Another sequence $x_3[n]$ is derived by taking 4-point inverse DFT of $X_3(k) = X_1(k)X_2(k)$. The value of $x_3[2]$ is _____.

3. Two sequences [a, b, c] and [A, B, C] are related as $\begin{bmatrix} A \\ B \\ C \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3^{-1} & W_3^{-2} \\ 1 & W_3^{-2} & W_3^{-4} \end{bmatrix} \begin{bmatrix} a \\ b \\ c \end{bmatrix}$ where $W_3 = e^{j\frac{2\pi}{3}}$.
If another sequence [p, q, r] is derived as,

$$\begin{bmatrix} p \\ q \\ r \end{bmatrix} = \begin{bmatrix} 1 & 1 & 1 \\ 1 & W_3^1 & W_3^2 \\ 1 & W_3^2 & W_3^4 \end{bmatrix} \begin{bmatrix} 1 & 0 & 0 \\ 0 & W_3^2 & 0 \\ 0 & 0 & W_3^4 \end{bmatrix} \begin{bmatrix} A/3 \\ B/3 \\ C/3 \end{bmatrix}$$

then the relationship between the sequences [p, q, r] and [a, b, c] is
(A) [p, q, r] = [b, a, c]
(B) [p, q, r] = [b, c, a]
(C) [p, q, r] = [c, a, b]
(D) [p, q, r] = [c, b, a]

4. Consider a discrete time periodic signal $x[n] = \sin\left[\frac{\pi n}{5}\right]$. Let a_k be the complex Fourier series coefficients of x[n]. The coefficients $\{a_k\}$ are non-zero when $k = Bm \pm 1$, where m is an integer. What is the value of B?

5. The N-point DFT of a sequence x[n]; $0 \le n \le N-1$ is given by:

$$X[k] = \frac{1}{\sqrt{N}} \sum_{n=0}^{N-1} x[n] e^{-j\frac{2\pi}{N}nk}, 0 \le k \le N-1$$

Denote this relation as X = DFT(x). For N= 4, which one of the following sequences satisfies DFT [DFT(x)] = x.
(A) x = [1 2 3 4]
(B) x = [1 2 3 2]
(C) x = [1 3 2 2]
(D) x = [1 2 2 3]

6. The DFT of a vector [a b c d] is the vector $[\alpha\ \beta\ \gamma\ \delta]$. Consider the product:

$$[p\ \ q\ \ r\ \ s] = [a\ \ b\ \ c\ \ d] \begin{bmatrix} a & b & c & d \\ d & a & b & c \\ c & d & a & b \\ b & c & d & a \end{bmatrix}$$

The DFT of the vector [p q r s] is the scaled version of:
(A) $[\alpha^2\ \ \beta^2\ \ \gamma^2\ \ \delta^2]$
(B) $[\sqrt{\alpha}\ \ \sqrt{\beta}\ \ \sqrt{\gamma}\ \ \sqrt{\delta}]$
(C) $[\alpha+\beta\ \ \beta+\delta\ \ \delta+\gamma\ \ \gamma+\alpha]$
(D) $[\alpha\ \ \beta\ \ \gamma\ \ \delta]$

7. The first six points of the 8-point DFT of a real valued sequence are 5, 1 – j3, 0, 3 – j4, 0 and 3 + j4. The last two points of the DFT are respectively:
(A) 0, 1 – j3
(B) 0, 1 + j3
(C) 1 + j3, 5
(D) 1 –j3, 5

8. A system with transfer function H(z) has impulse response h(n) defined as h(2) = 1, h(3) = –1, and h(k) = 0 otherwise. Consider the following statements.
S1: H(z) is a low-pass filter.
S2: H(z) is an FIR filter.
Which of the following is correct?
(A) Only S2 is true.
(B) Both S1 and S2 are false.
(C) Both S1 and S2 are true, and S2 is reason for S1.
(D) Both S1 and S2 are true, but S2 is not reason for S1.

9. The 4-point Discrete Fourier Transform (DFT) of a discrete time sequence (1, 0, 2, 3) is:
(A) [0, –2 + 2j, 2, –2 –2j]
(B) [2, 2 + 2j, 6, –2 –2j]
(C) [6, 1 –3j, 2, 1 + 3j]
(D) [6, –1 + 3j, 0, –1 –3j]

10. {x(n)} is a real-valued periodic sequence with a period N. x(n) and X(k) form N-point Discrete Fourier Transform (DFT) pairs. The DFT Y(k) of the sequence $y(n) = \frac{1}{N}\sum_{r=0}^{N-1} x(r)x(n+r)$ is:
(A) $|X(k)|^2$
(B) $\frac{1}{N}\sum_{r=0}^{N-1} X(r)X^*(k+r)$
(C) $\frac{1}{N}\sum_{r=0}^{N-1} X(k+r)X(r)$
(D) 0

Answer Key

Level I

1. (B)	**2.** (A)	**3.** (D)	**4.** (B)	**5.** (C)	**6.** (A)	**7.** (C)	**8.** (B)	**9.** (A)	**10.** (D)
11. (C)	**12.** (D)	**13.** (A)	**14.** (B)	**15.** (C)	**16.** (D)	**17.** (A)			

Level II

1. (6)	**2.** 11	**3.** (C)	**4.** (10)	**5.** (B)	**6.** (A)	**7.** (B)	**8.** (A)	**9.** (D)	**10.** (A)